ML BUBBLE 2026 · TE-BE DESIGN & SOLVE

# Predicting 30-Day Hospital Readmission

*Turning 101,766 discharge records into a ranked call list for a follow-up team that can only reach one patient in five.*

| 40.2% | 2.05× | 0.238 |
|---|---|---|
| of readmissions caught | lift over random calling | PR-AUC vs 0.113 base rate |

*Held-out test set · 19,773 encounters · 13,998 patients unseen in training*

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THE PROBLEM

# A capacity problem, not a prediction problem

*“We can call 20% of today’s discharges. Which 20%?”*

- About 1 in 9 diabetic discharges returns within 30 days — many are preventable: a phone call, a medication check, an earlier appointment.
- A discharge-planning team can reach only a fraction of patients per day, so the useful question is which patients get that scarce attention.
- The team works down a ranked list — ranking quality matters more than absolute probability, and the threshold comes from budget, not an arbitrary 0.5 cutoff.

**Consequence:** the model is optimised for PR-AUC and recall-at-capacity — never accuracy.

THE DATA

## Diabetes 130-US Hospitals, 1999–2008

*UCI ML Repository dataset 296 · 10 years across 130 hospitals*

| | |
|---|---|
| **101,766** raw encounters | **71,518** distinct patients |
| **99,340** encounters modelled | **11.4%** readmitted <30 days |

**Binarised** <30 vs everything else • **2,423 death/hospice** discharges removed • '?' kept as an **explicit Unknown** category — missingness carries signal • 39 engineered features survive to modelling.

METHOD

# From raw encounter to ranked call list — without cheating

1

### Clean

Remove death/hospice discharges. Keep '?' as an explicit category.

2

### Engineer

2,000+ ICD-9 codes → 9 chapters. Utilisation & regimen-churn features.

3

### Split

GroupShuffleSplit on patient_nbr. Train/val/test = 63.6k / 16.0k / 19.8k.

4

### Compare

6 models, 5-fold GroupKFold. Winner picked on validation PR-AUC.

5

### Calibrate

Platt scaling, then a threshold set from follow-up capacity.

***The test set is read exactly once, at the very end.*** *Model choice, hyperparameters, calibration and threshold are all fixed against validation.*

THE TRAP

## 99,340 encounters belong to only 69,987 patients — one appears 40 times

**✕ Split by row**

Same patient lands in train and test — the model memorises individuals instead of learning transferable risk. Every metric comes out inflated.

**✓ Split by patient_nbr**

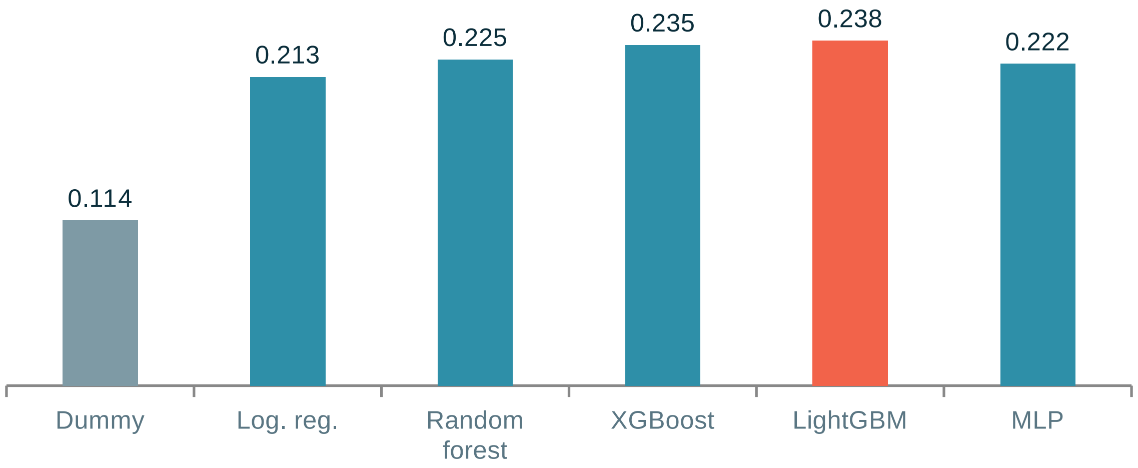
GroupShuffleSplit for train/val/test, GroupKFold for every CV fold. No patient can appear on both sides — enforced by a test, not convention.

`test_split_never_puts_a_patient_on_both_sides` — a materially higher ROC-AUC on this dataset almost always means a leaked split, not a better model. Published results sit at 0.64–0.70.

# Six models, one selection rule — chosen before the test set was read

*5-fold GroupKFold on training patients for selection; a single held-out test evaluation at the end.*

### Test PR-AUC  (base rate 0.113)




**Selected: LightGBM — Test PR-AUC 0.238 · ROC-AUC 0.684 · Recall 40.2% at 20% capacity.**

**Which gaps are real?** Paired bootstrap on Δ PR-AUC — vs XGBoost: +0.0022 [−0.0017, +0.0058], not resolved (a coin flip). vs Random forest / Logistic reg.: resolved in LightGBM's favour. Boosting beats bagging and linear models; which booster wins is not resolved by this data.

### CLASS IMBALANCE — 11.4% positives, handled explicitly

*Left alone, every model collapses to 'no readmission' for everyone — 88.6% accuracy, catches nothing.*

| | |
|---|---|
| **Logistic reg.** | balanced |
| **Random forest** | balanced |
| **XGBoost** | ~7.8× |
| **LightGBM** | balanced |
| **MLP** | unweighted — judged on ranking |

**SMOTE was measured, not assumed away.** Oversampling inside each CV fold scored 0.2194 PR-AUC vs 0.2200 for class weighting — no better, and it adds a resampling step to serving. Kept on the evidence.

**The cost of class weighting.** It deliberately distorts predicted probabilities upward — fine for ranking, wrong for a number a clinician reads. That's why calibration is a separate step (next).

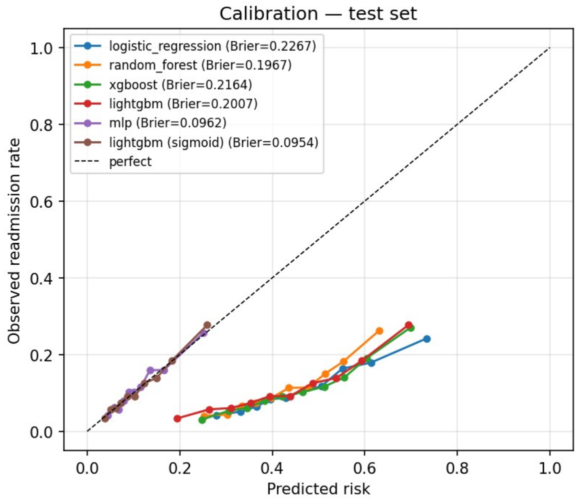
# 78.2% accurate. So is doing nothing — at 88.6%.

| 78.2% | 88.6% | 88.8% |
|---|---|---|
| our model | “nobody is readmitted” | best accuracy at any threshold |
| 40.2% of readmissions caught | 0% caught — flags no one | flags 0.4% of patients |

**With an 11% positive rate, accuracy rewards inaction — nothing here was selected using it.** The deployed threshold is 0.166, not 0.5: at 0.5 the calibrated model flags zero patients.

## CALIBRATION — making the number mean what it says

*Class weighting is what makes ranking work and probabilities wrong. Platt scaling fixes the probabilities without touching the ranking.*

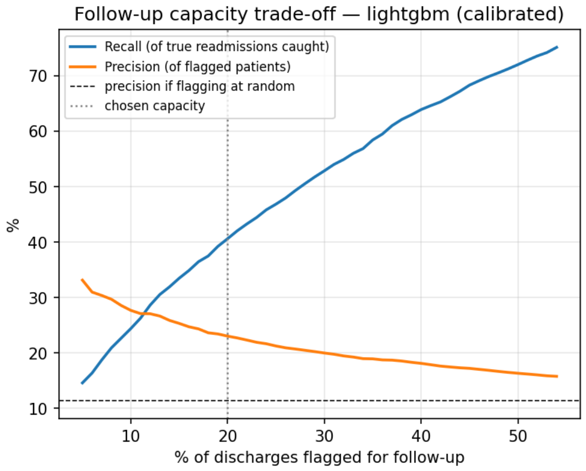



| | Brier | ROC-AUC | PR-AUC |
|---|---|---|---|
| LightGBM, raw | 0.2007 | 0.6839 | 0.2376 |
| + Platt scaling | 0.0954 | 0.6839 | 0.2376 |

**Brier more than halves; discrimination is bit-identical.** Platt scaling is strictly monotonic — it cannot reorder a single patient. The call list is unchanged while a predicted 18% now means roughly 18%.

# The threshold is a budget decision — and acting on it beats the alternatives

## Precision & recall vs. follow-up call capacity




| Capacity | Recall | Precision |
|---|---|---|
| 20% ← deployed | 40.6% | 23.1% |
| 40% | 63.9% | 18.1% |

**Doubling capacity from 20% to 40% buys 23 points of recall for 5 points of precision.** At 20%: 901 true positives, 2,965 false alarms, 1,343 missed. Whether the trade is worth taking is a hospital finance question — the model supports either answer without retraining.

## CLINICAL VALUE — does acting on it beat what they do today?

*Decision curve analysis: pt is the risk at which a call becomes worthwhile — it encodes the cost ratio of a call vs. a missed readmission.*

| +0.0157 | −0.0617 | 0.0000 |
|---|---|---|
| Use the model | Call everyone | Call nobody |

*Net benefit*

**15.6 extra readmissions caught per 1,000 discharges, net of the false alarms they cost.** The model beats both alternatives for pt between 0.025 and 0.43 — every realistic hospital sits inside that range. Note ‘call everyone’ is worse than doing nothing at this exchange rate: false-alarm burden swamps the benefit.

# What the model reads — and whether the number holds up

*Permutation importance and SHAP agree on the top of the list.*

1. **Prior inpatient admissions**

   Utilisation history beats anything measured during the stay.

2. **Where the patient is discharged to**

   Home lowers risk; a care facility raises it — a proxy for frailty.

3. **Depth of prior encounter history**

   How often this patient has been through the system before.

4. **Primary diagnosis group**

   Circulatory pushes risk up; respiratory pushes it down.

**Nothing suspicious at the top — no identifier, no date artefact.**

## ROBUSTNESS — is 0.684 real?

Across 7 random seeds, the pipeline averages **PR-AUC 0.224 ± 0.010** and **ROC-AUC 0.678 ± 0.006**. 95% CI from 2,000 cluster resamples of patients:

```
ROC-AUC 0.684 [0.671, 0.698]   ·   PR-AUC 0.238 [0.213, 0.265]   ·
Recall 40.2% [37.6%, 42.7%]
```

## ABLATION — seven approaches tried, none beat the baseline

| Variant | CV PR-AUC | Δ |
|---|---|---|
| **A — current, 39 features** | 0.2200 | baseline |
| **F — soft-vote RF+XGB+LGBM** | 0.2191 | −0.0009 |
| **G — SMOTE, not weighting** | 0.2194 | −0.0006 |

**Every delta is inside the ±0.0069 fold-to-fold noise — no measurable difference.** Seven approaches converging on 0.22 is itself the finding. That’s a ceiling on method — the learning curve (still rising, +0.0049 from the last 43% of data) shows it isn’t a ceiling on data.

# The subgroup gap that would block deployment

*Per-subgroup performance at the deployed threshold. Groups under 200 test encounters are suppressed.*

| Age band | Flag rate | Recall | ROC-AUC |
|---|---|---|---|
| **Under 40** | 20.2% | 58.0% | 0.784 |
| **40–60** | 14.8% | 39.6% | 0.712 |
| **60–80** | 20.8% | 39.7% | 0.673 |
| **80 and over** | **23.0%** | **35.8%** | **0.613** |

**Race and gender gaps are modest — age is the problem.** The two large racial groups get near-identical flag rates and recall. Patients over 80 are flagged most often (23.0%) while the model ranks them worst (ROC-AUC 0.613) — the tool spends disproportionate follow-up capacity where its ordering is least trustworthy. This is not fixable by tuning.

**EQUITY EXPERIMENT — we tried the obvious fix**

One capacity threshold per age band. Recall drops 40.2% → 39.7% (901 → 891 true positives); 80+ ROC-AUC is unchanged at 0.613 under both policies. **Thresholds move capacity — they cannot change ranking.**

**ROOT CAUSE — a data limit, not a modelling one**

Every feature's signal decays with age — an 80+ patient carries about a third the separable signal of a <40 patient. A dedicated 80+ model is worse (0.588 vs 0.613): exposure to younger patients helps rank older ones. **Closing this needs frailty and functional-status data this dataset doesn't carry.**

# Decision support, not autonomous action

*The model produces a ranked call list. A clinician decides what to do with it.*

## ARCHITECTURE

**Nightly batch** — The discharge list becomes a ranked worklist, written straight into the follow-up team's queue.

**Real-time endpoint** — FastAPI /score returns risk, flag, band and disclaimer. Sub-millisecond tree inference.

**One 5 MB artefact** — Preprocessing, booster, calibrator and threshold travel together in a single joblib file.

**Two serving bugs offline metrics could never have caught:**

• Feature engineering required the readmitted label — which doesn't exist at discharge.

• prior_encounters was counted per-batch — a single-row request always came out zero, killing the model's strongest feature.

*Monitor flag rate, realised-vs-predicted readmission rate, calibration drift and subgroup gaps — on a schedule, not once at launch.*

## WHAT WE'D SAY TO A HOSPITAL — honest about the ceiling

- **Discrimination is modest — and that's a ceiling on method.** 0.684 sits inside the 0.64–0.70 band published for this dataset. Seven approaches converged there.
- **Insurance coding is not physiology.** payer_code_Unknown is a top-6 driver — it may not transfer to a new hospital; first thing to re-examine on local data.
- **The data is 1999–2008.** Coding standards, drug availability and discharge policy have all moved. Recalibration is mandatory.
- **The oldest patients cannot be ranked well — and we know why.** Every feature's signal decays with age. Three fixes failed; it needs frailty data this dataset lacks.

`github.com/sgoel2be24-cyber/hospital-readmission-risk` `make setup && make data && make train` · 27 tests · ~60s end to end